

MT – 101- Practise & Assignment Questions

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Do all the questions marked with a * (Asterisk) as an assignment and submit by Oct 1, 26.

SETS

1. There are 35 students in art class and 57 students in dance class. Find the number of students who are either in art class or in dance class. When two classes meet at different hours and 12 students are enrolled in both activities.
2. Each student in a class of 40 plays at least one indoor game chess, carrom and scrabble. 18 play chess, 20 play scrabble and 27 play carrom. 7 play both chess and scrabble, 12 play both scrabble and carrom and 4 play all chess, carrom and scrabble. Find the number of students who play (i) chess and carrom. (ii) chess, carrom but not scrabble.
3. In a survey among 140 students, 60 likes to play videogames, 70 likes to play indoor games, 75 likes to play outdoor games, 30 play indoor and outdoor games, 18 like to play video games and outdoor games, 42 play video games and indoor games and 8 likes to play all types of games. Use the Venn diagram to find
 - (i) Students who play only outdoor games
 - (ii) Students who play video games and indoor games, but not outdoor games
4. * A construction site is divided into three activity zones: A (Excavation), B (Foundation), and C (Steelwork). A survey of 50 workers show that: 18 are working in Zone A, 22 in Zone B, 25 in Zone C. Similarly, 8 are in both A and B, 9 in both A and C, 10 in both B and C, 5 in all three zones.
 - (i) Using the Principle of Inclusion-Exclusion, find how many workers are part of the union of the three zones?
 - (ii) How many are not assigned to any zone? Show it using a Venn diagram
5. *A survey of 100 students asks which sports they play. Fifty-five students play cricket, 45 play football, and 40 play badminton. Twenty students play both cricket and football, 18 play both cricket and badminton, and 15 play both football and badminton. Ten students play all three sports. Find
 - a) The number of students who play exactly two sports.
 - b) The number who play only cricket.
 - c) The number who play only football,
 - d) The number who play atleast one of the three sports.
 - e) The number who play none of these sports.
 - f) Draw a Venn diagram and fill all regions.

Probability

1. A survey of 100 students records how they travel to university. 45 students use a bus, 30 use a car, and 15 use both a bus and a car. Find the relative frequency of students who use a bus, the probability that a randomly selected student uses

a bus or a car; the probability that a randomly selected student uses neither a bus nor a car. State whether the probabilities calculated above are empirical or classical probabilities.

2. Two cards are drawn without replacement from a standard deck of 52 cards. Find the probability that atleast one of the two cards is a heart or diamond.
3. *Two fair dice are rolled simultaneously. Find the probability that:
 - (a) the sum is a prime number,
 - (b) the sum is greater than 7
 - (c) the numbers are different and their sum is even,
 - (d) the sum is 6.

Q4. * A factory records the following results from 1,200 products:

Machine	Defective	Non-defective
A	18	582
B	30	570

Using relative frequency, find:

- (a) the probability that a randomly selected product is defective,
- (b) the probability that it is non-defective,
- (c) if 5,000 products are produced under similar conditions, estimate the number expected to be defective,
- (d) estimate the number expected to be non-defective.

Q5. A basketball player takes 200 shots and successfully scores 128 times. Using relative frequency, find:

- (a) the probability of scoring,
- (b) the probability of missing,
- (c) the expected number of successful shots in 500 attempts,
- (d) the expected number of misses in 750 attempts.

Permutations & Combinations

1. A committee of 4 students is to be selected from 6 boys and 5 girls. Find the number of committees containing at least 2 girls. Find the number of committees containing exactly 2 girls.
2. * Six students A, B, C, D, E, and F are standing in a row. In how many ways can they be arranged if A and B must stand together? In how many ways can they be arranged if A and B must not stand together?

3. From 8 students, three positions of Captain, Vice-Captain, and Secretary are to be filled. In how many ways can the three positions be assigned? If A must be Captain, in how many ways can the remaining two positions be assigned?
4. *The word COMPUTER contains 8 different letters. In how many ways can 5 letters be selected and arranged from the word? In how many of these arrangements does the letter C appear first?

Area & Perimeter

1. * A university is planning a research facility made of a rectangle with a semicircular lab attached to one shorter side and a triangular room attached to the other shorter side. The rectangular hall is 10 m long and 8 m wide. The triangular room has a base of 8 m, height of 6 m, and two equal sides of 10 m each. Find the total area of the complete structure and the outer perimeter of the complete structure.
2. * A university is designing a garden made from a rectangle with a semicircular section attached to one of its shorter sides. The rectangular part is 18 m long and 10 m wide. Find the total area of the garden. Find the outer perimeter of the garden. Draw a labelled sketch of the compound figure.

For questions related to Percentages, see chap 3 of the given book